

PUBUDU GUNASEKARA

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MS Computer Science (Northeastern) • Seeking Software Engineering Co-op: January – August 2027

EDUCATION

Northeastern University, Khoury College of Computer Sciences — San Jose, CA

M.S. in Computer Science | GPA: 4.0 / 4.0

Jan 2026 – Present

Expected 2028

Conestoga College — Waterloo, ON, Canada

Postgraduate Certificate, Mobile Solutions Development | GPA: 3.74 / 4.0

2022 – 2023

Victoria University — Melbourne, Australia

Bachelor of Information Technology (Distinction)

2018 – 2021

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL, C++

Backend & Distributed: Spring Boot, Node.js / Express, Apache Kafka, Redis, PostgreSQL, MongoDB, REST APIs, microservices, hexagonal architecture

Infra & DevOps: Docker, Docker Compose, GitHub Actions, Jenkins, CI/CD, Prometheus, Grafana, AWS

Frontend: React, Next.js, React Native, Vite, Redux

EXPERIENCE

Gunasekara Transport — Web Developer

Dec 2025 – Present · Remote

- Building the corporate website and web app interfaces, plus an internal business management system (HR, operational workflows, and reporting) that replaces manual processes, using React.js, JavaScript, PHP, WordPress, and Docker.

Virtusa — Associate Engineer – QA

Jun 2021 – Mar 2022

Colombo, Sri Lanka

- Built Java and Selenium WebDriver test automation for a British Telecommunications platform, designing and executing test cases and test plans to verify system functionality across agile sprints.
- Led test planning and ran client demos and daily stand-ups for the module under the QA lead's delegation, and collaborated with developers to validate bug fixes.
- Tracked QA metrics including defect densities and open defect counts, working across CST Environment, Camunda Cockpit, RabbitMQ, Swagger UI, and MQ Console.

PROJECTS

Distributed Task Scheduler

Java 21, Spring Boot 3.5, Apache Kafka, Redis, PostgreSQL, Prometheus, Grafana, Docker, GitHub Actions • [Code](#)

- Built a distributed task scheduler on a hexagonal architecture enforced at build time via ArchUnit, with at-least-once delivery over Kafka and a state-machine guard that prevents duplicate execution on redelivery.
- Implemented a Redis distributed lock (atomic SETNX acquire, Lua compare-and-delete release) and an exponential-backoff retry scheduler (10s → 30s → 90s) with dead-letter handling.
- Instrumented the system end-to-end with Prometheus metrics and Grafana dashboards; containerized with Docker Compose and automated build/test in GitHub Actions CI.

AI Code Review Assistant

React 19 / Vite, Node.js / Express, PostgreSQL, Redis, gpt-4o-mini, Docker, AWS (EC2 + Amplify) • [Code](#) · [Live Demo](#)

- Built and deployed a full-stack app that posts structured gpt-4o-mini code reviews on GitHub pull requests via GitHub OAuth, returning severity- and category-tagged findings in a strict JSON schema.
- Engineered per-user Redis rate limiting with a fail-open policy and standard rate-limit headers, plus diff truncation to stay within the model's context budget.

Travel Day Scheduler

Python, Weighted A* Search • Graduate algorithms project (team) • [Code](#)

- Implemented a weighted / anytime A* itinerary planner with iterative weight escalation and heap pruning to bound memory under large search spaces; built the route-visualization frontend.